HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F70LB

Life Insurance Mathematics B

Excel Test

April Examination and Assessment Diet 2024-2025

Duration: 2 Hours 30 minutes

Please answer all questions in Part A and Part B of this paper.

Total 90 marks

Important Instructions – PLEASE READ

You have been supplied with two answer booklets in which to complete your work. One is an Excel spreadsheet and the other is a Word document. Note that Part B is a written exercise which depends on Part A, so you will need to attempt Part A before proceeding to Part B.

Submit your answers to Canvas as two files:

- *Part A: Completed "Part A Excel Answer Booklet.xlsx"*
- *Part B: Completed "Part B Word Answer Booklet.docx"*

Mark scheme:

- *Part A (two questions **72 marks**): Marks are awarded for correct answers and working. There are no marks directly awarded for how you set out your spreadsheets. However, it is more likely to be possible to award marks for working if your spreadsheet is clear and easy to follow.*
- *Part B (one question **18 marks**): In part B, you will be asked to write about your answers from Part A.*

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Part A

Question 1

The sheet *"Q1 Base"* of your Excel answer template shows the force of mortality for lives aged 40 to 60 at intervals of 0.1 years.

In your answer template, do your calculations in *"Q1 Working"* then link (not copy) your answers to the sheet *"Q1 Answers".*

In all the following calculations use a step size of $h = 0.1$ years.

(a) Use an Euler Scheme (Euler's Method) to calculate ${}_tp_{40}$ for each $t = 0, 0.1, 0.2, \ldots, 19.9, 20$ Enter the value of ${}_{20}p_{40}$ in your *"Q1 Answers"* sheet.

(5 marks)

A 20-year term assurance is issued to a life age 40. The office charges a gross premium of £180 continuously. A sum assured of £100,000 is paid immediately on death within the term and there is no maturity benefit. The office uses a force of interest of $\delta = 0.07$ and the same forces of mortality as part (i). Assume that expenses are zero.

(b) Use an Euler Scheme (Euler's Method) to calculate the **gross** premium policy value $V(t)$ for each $t = 0, 0.1, 0.2, \ldots, 19.9, 20$. Enter (as a link) the value of $V(0)$ in your *"Q1 Answers"* sheet.

(7 marks)

(c) Use an Euler Scheme (Euler's Method) to calculate the **net** premium policy value $V(t)$ for each $t = 0, 0.1, 0.2, \ldots, 19.9, 20$. Enter (as a link) the value of the net premium in your *"Q1 Answers"* sheet.

(HINT: You might find it easiest to make a copy of your answer to (b) than to start again.)

(3 marks)

(d) Now suppose that the sum assured starts at £200,000 at age 40 and falls linearly to £0 at the end of the 20-year term. Use an Euler Scheme (Euler's Method) to calculate the **gross** premium policy value $V(t)$ for each $t = 0, 0.1, 0.2, \ldots, 19.9, 20$. Enter (as a link) the value of $V(0)$ in your *"Q1 Answers"* sheet.

(HINT: You might find it easiest to make a copy of your answer to (b) than to start again.)

(5 marks)

QUESTION CONTINUES ON THE NEXT PAGE

QUESTION CONTINUED FROM THE PREVIOUS PAGE

(e) Calculate the following using your answer to part (a) and a force of interest of $\delta = 0.07$:

(i) The expected present value of a 20-year term assurance of £1 paid instantly issued to a life aged 40. Do your calculation using the trapezium rule approximation as follows:

$$\bar{A}^{\,1}_{40\,\overline{:20}|} = \int_0^{20} v^s \,{}_sp_{40}\,\mu_{40+s}\,ds \approx \sum_{k=0}^{200} 0.1 w_k\, v^{0.1k}\,{}_{0.1k}p_{40}\,\mu_{40+0.1k}$$

where

$$w_k = \begin{cases} 0.5, & \text{if } k = 0 \text{ or } k = 200, \\ 1, & \text{otherwise.} \end{cases}$$

Enter (as a link) the answer in your *"Q1 Answers"* sheet.

(ii) Using the same method as part (e)(i), calculate the expected present value of a 20-year annuity of £1 paid continuously issued to a life aged 40. Enter (as a link) the answer in your *"Q1 Answers"* sheet.

(iii) Using your answers to (e)(i) and (e)(ii), calculate the gross premium policy value $V(0)$ at time zero of a 20-year term assurance issued to a life age 40. The gross premium is £180 a year paid continuously. A flat sum assured of £100,000 is paid immediately on death. Enter (as a link) the value of $V(0)$ in your *"Q1 Answers"* sheet.

(9 marks)

[Total 29 marks]

PLEASE TURN OVER

Question 2

A life insurance company issues a 30-year policy to a life age 70. The policy has the following features:

- It has a single premium of £140,000 paid at time zero by the policyholder to the insurer
- An annuity benefit of £10,000 paid at the end of every year to the policyholder
- A death benefit of £50,000 at the end of the year of death if death occurs within the 30-year term

The insurer makes the following assumptions:

- Initial expenses at time zero of £10,000
- Ongoing expenses of £100 at the start of each year except the first (so they are not paid at time 0 but they are paid at time 1,2,3,…)
- Interest on reserves and cashflows of 6% a year effective
- The risk discount rate is 9%

Create a profit-test to calculate the following:

(a) The expected present value of profit before allowing for reserves (22 marks)

(b) The expected present value of profit with allowance for reserves (12 marks)

(c) The profit margin with allowance for reserves (1 mark)

(d) The discounted payback period with allowance for reserves (4 marks)

(e) The internal rate of return with allowance for reserves (4 marks)

Enter (as a link) these values in your *"Q2 Answers"* sheet.

[Total 43 marks]

PLEASE TURN OVER

Part B

Question 1

You are in an actuarial graduate role and you have spent your afternoon working on the calculations in Part A Question 1. You were given those calculations to do by your line manager, Alex, who is a recently qualified actuary.

Alex is not in the office and you are about to finish for the day. Alex has asked you to write her a quick email before you leave to tell her what you did on Question 1, what answers you got and whether you think the answers are reasonable.

In your email, you should include:

- A brief statement of your answers to Question 1 parts (b), (c), (d) and (e). (Say if you did not finish any part.)

- A discussion of how your answers compare to each other. Say whether your answers seem reasonable to you and why.

- Comments on anything that will help Alex to review your calculations from Question 1. This could include mentioning anything that is unfinished or which you think may not be correct.

(12 marks)

A further 6 marks are available for writing and structure of your email.

You should write no more than 500 words.

[Total 18 marks]

END OF PAPER